Analytic geometry

1. Planes in space $\mathbb{R}^3$

Fix a cartesian frame in space with origin O and triad $\{\vec{i}, \vec{j}, \vec{k}\}$. Every point $M(x,y,z)$ has position vector $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$.

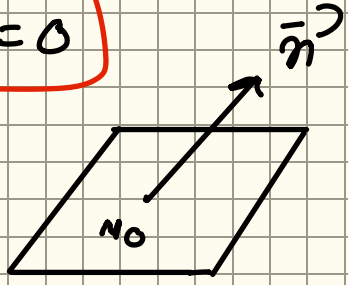
Def. For $M_0(x_0, y_0, z_0)$ with $r_0 = x_0\vec{i} + y_0\vec{j} + z_0\vec{k}$, the parallelogram rule gives us:

$$\overrightarrow{MM_0} = \vec{r}_0 - \vec{r} = (x_0 - x)\vec{i} + (y_0 - y)\vec{j} + (z_0 - z)\vec{k}$$

1.1. Plane through a point with normal vector:

Def. Let $M_0(x_0, y_0, z_0)$ and $\vec{n} = a\vec{i} + b\vec{j} + c\vec{k} \neq \vec{0}$. Let P be the plane through M_0 perpendicular to $\vec{n}$. For $M(x,y,z) \in P$:

$$\vec{r} - \vec{r}_0 \perp \vec{n} \iff \vec{n} \cdot (\vec{r} - \vec{r}_0) = 0 \iff$$
$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$



Obs. Setting $d = -ax_0 - by_0 - cz_0$:

$$ax + by + cz + d = 0$$

This is the general equation of a plane; $\vec{n} = a\vec{i} + b\vec{j} + c\vec{k}$ is normal to it.

ex. The plane xOy passes through O with normal $\vec{k}$: take $a = b = 0$, $c = 1$. Its equation is $z = 0$.

1.2 Plane through three points:

Def. Let $M_i(x_i, y_i, z_i)$, $i = 1,2,3$, be non-collinear and P the plane through them. For $M(x,y,z) \in P$, the four points are coplanar:

$$\begin{vmatrix} 1 & x & y & z \\ 1 & x_1 & y_1 & z_1 \\ 1 & x_2 & y_2 & z_2 \\ 1 & x_3 & y_3 & z_3 \end{vmatrix} = 0$$

1.3. Plane through a point and two dir. vectors:

Def. Let P pass through $M_0(x_0, y_0, z_0)$, parallel to non-collinear $\vec{v}_i = x_i\vec{i} + y_i\vec{j} + z_i\vec{k}$, $i = 1,2$. For $M \in P$, the vectors $\vec{r} - \vec{r}_0$, $\vec{v}_1$, $\vec{v}_2$ are coplanar $\Leftrightarrow$ their triple product vanishes (only area, no volume):

$$\begin{vmatrix} x - x_0 & y - y_0 & z - z_0 \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix} = 0$$

1.4. The pencil of planes through a line:

Def. Every plane through the intersection line of:

(1) $a_1x + b_1y + c_1z + d_1 = 0$, (2) $a_2x + b_2y + c_2z + d_2 = 0$

has the form:

(3) $a_1x + b_1y + c_1z + d_1 + \lambda(a_2x + b_2y + c_2z + d_2) = 0$

2. Straight lines in space:

2.1. Direction ratios and direction cosines:

Def. A direction in space is fixed by $\vec{v} = l\vec{i} + m\vec{j} + n\vec{k} \neq 0$. (l, m, n) are its direction ratios (any proportional triple works as well).

Obs. If $\|\vec{v}\| = 1$, then $l^2 + m^2 + n^2 = 1$ and $l = \vec{v} \cdot \vec{i} = \cos\alpha$, $m = \cos\beta$, $n = \cos\gamma$, where α, β, γ are the angles to the axes - the direction cosines:

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

2.2. Line through a point with a direction vector:

Def. Let d pass through $M_0(x_0, y_0, z_0)$ with direction $\vec{v} = l\vec{i} + m\vec{j} + n\vec{k}$. For $M(x, y, z) \in d$: $\vec{r} - \vec{r}_0 = t\vec{v}$, $t \in \mathbb{R}$:

$$x = x_0 + lt, \quad y = y_0 + mt, \quad z = z_0 + nt$$

$$\Rightarrow \boxed{\frac{x - x_0}{l} = \frac{y - y_0}{m} = \frac{z - z_0}{n} = t}$$

ex. Take $M_0 = O$, $\vec{v} = \vec{i}$: so $x_0 = y_0 = z_0 = 0$, $l = 1$, $m = n = 0$:

$$\frac{x - 0}{1} = \frac{y}{0} = \frac{z}{0} \Leftrightarrow \begin{cases} y = 0 \\ z = 0 \end{cases}$$

2.3. Line through two points:

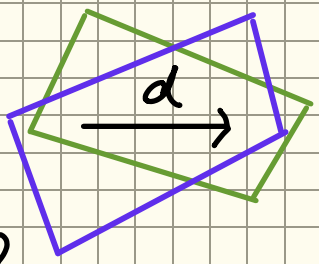
Def. Let $M_0(x_0, y_0, z_0)$, $M_1(x_1, y_1, z_1)$. The line is determined by M_0 and direction $\vec{r}_1 - \vec{r}_0$, giving direction ratios: $(x_1 - x_0, y_1 - y_0, z_1 - z_0)$:

$$\frac{x-x_0}{x_1-x_0} = \frac{y-y_0}{y_1-y_0} = \frac{z-z_0}{z_1-z_0}$$

2.4. Line as the intersection of two planes:

Def. Let $d = P_1 \cap P_2$:

$$\begin{cases} a_1x + b_1y + c_1z + d_1 = 0 \\ a_2x + b_2y + c_2z + d_2 = 0 \end{cases}$$



with normals: $\vec{n_1} = a_1\vec{i} + b_1\vec{j} + c_1\vec{k}$

$\vec{n_2} = a_2\vec{i} + b_2\vec{j} + c_2\vec{k}$

Since $\vec{n_1}, \vec{n_2} \perp d \Rightarrow d \parallel \vec{n_1} \times \vec{n_2} \Rightarrow$ the direction ration for line d is:

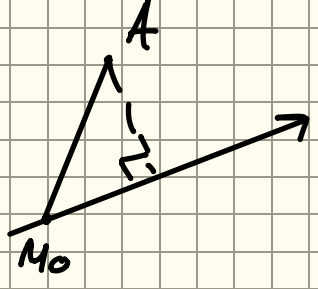
$$\left(e = \begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix}, \; m = \begin{vmatrix} c_1 & a_1 \\ c_2 & a_2 \end{vmatrix}, \; n = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \right)$$

3. Distance problems:

3.1. Distance from point to line:

Def. Let d pass through M_0 with dir $\vec{v}$ and let A be a point. $dist(A,d)$ is the height of the parallelogram spanned by $\vec{v}$ and $\overrightarrow{M_0A}$.

$$dist(A,d) = \frac{\|\vec{v} \times \overrightarrow{M_0A}\|}{\|\vec{v}\|}$$



3.2. Distance from point to plane:

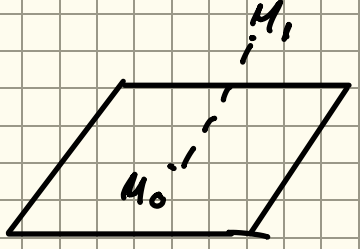
Def. Let $P: ax + by + cz + d = 0$ and $M_0(x_0, y_0, z_0)$;

Let $M_1(x_1, y_1, z_1)$ be the orthogonal projection of M_0 onto P s.t. $ax_1 + by_1 + cz_1 + d_1 = 0$. The normal $(n\ \vec{n} = a\vec{i} + b\vec{j} + c\vec{k}$ and $\overrightarrow{M_1M_0} = (x_0 - x_1)\vec{i} + (y_0 - y_1)\vec{j} + (z_0 - z_1)\vec{k}$:

$$\vec{n} \cdot \overrightarrow{M_1M_0} = a(x_0 - x_1) + b(y_0 - y_1) + c(z_0 - z_1) = ax_0 + bx_0 + cx_0 + d$$

$$|\vec{n} \cdot \overrightarrow{M_1M_0}| = |\vec{n}| \cdot |\overrightarrow{M_1M_0}| \cdot \cos 0 = \sqrt{a^2+b^2+c^2} \cdot dist(M_0, P) \Rightarrow$$

$$\boxed{dist(M_0, P) = \frac{|ax_0 + by_0 + cz_0 + d_0|}{\sqrt{a^2+b^2+c^2}}}$$



Exercises:

$P: 7(x-1) + 2(y-2) + z = 0$

$\vec{n} = 7\vec{i} + 2\vec{j} + \vec{k} \perp P$

$M_0(1, 2, 0) \in P$

$P_2: x - y + z - 5 = 0$, $\vec{n} = \vec{i} - \vec{j} + \vec{k} \perp P$

$M(8, 4, 1) \in P_2$

ex. $P_1: x - y + z - 1 = 0$ $\quad d = P_1 \cap P_2$

$P_2: x + y + z + 3 = 0$

$$\begin{cases} x-y+z=1 \\ x+y+z=-3 \end{cases} \qquad \vec{n_1} = \vec{i} - \vec{j} + \vec{k} \perp P_1$$
$$\vec{n_2} = \vec{i} + \vec{j} + \vec{k} \perp P_2$$

$$\bar{A} = \left[\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 1 & 1 & 1 & -3 \end{array}\right] \Rightarrow \operatorname{rang} A = \operatorname{rang} \bar{A}$$

x, y - n.v.
$z = \alpha \in \mathbb{R}$ $\Rightarrow$

$$x - y = 1 - \alpha$$
$$x + y = -3 - \alpha$$
$$2x = -2 - 2\alpha \Rightarrow x = -1 - \alpha$$
$$y = -2 = -2 + 0 \cdot \alpha$$
$$z = \alpha = 0 + 1 \cdot \alpha$$

$M_0(1, -2, 0) \in d$

$\vec{v} = -\vec{i} + 0 \cdot \vec{j} + \vec{k} \parallel d$

$$\vec{n_1} \times \vec{n_2} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{vmatrix} = -2\vec{i} + 2\vec{k} = 2\vec{v}$$

$$\frac{x+1}{-1} = \frac{y+2}{0} = \frac{z}{-1}$$